

# HXC-108\_web\_evidence

## HXC-108 — HelixCode WEB client: Real Video-QA Recording (curated evidence)

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Run-id                       | HXC-108_web                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Date (UTC)                   | 2026-06-23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| HEAD at recording            | b04e9bf5 (binaries built at prior HEAD; server binary helix_code/bin/helixcode HelixCode Server v3.0.0, build 2026-06-20, commit 454319f1)                                                                                                                                                                                                                                                                                                                                                                                                      |
| Surface                      | HelixCode <b>WEB</b> client — the in-browser LLM Console (helix_code/web/frontend/index.html + static/app.js/app.css/logo.png) served by the Gin server (cmd/server → bin/helixcode) at Static("/", ". /web/frontend/index.html") + Static("/static", ". /web/frontend/static") (internal/server/server.go:369-370)                                                                                                                                                                                                                             |
| Capture method               | <b>headless Chrome via chromedp</b> (Go dep github.com/chromedp/chromedp v0.15.1, present in the inner module). Frames captured directly from the headless-browser viewport via chromedp.CaptureScreenshot → ffmpeg H.264 +faststart yuv420p. <b>TCC-free + window-scoped by construction</b> (§11.4.154/.159): chromedp captures ONLY the 1280×1400 browser viewport — never the host desktop — so it needs NO macOS Screen-Recording grant (unlike the screencapture window path that is TCC-blocked on this host, see the TUI/server slice). |
| Recordings dir               | /Volumes/T7/Downloads/Recordings/ (gitignored raw corpus, §11.4.128; outside the repo tree → inherently untracked)                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Prefix                       | helixcode (from HELIX_RELEASE_PREFIX in <repo-root>/ .env, §11.4.155/.151)                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Recording resource ownership | §11.4.119 — this agent was the SOLE owner of the server port (: 18080) + the headless browser for this slice; no desktop/terminal_ui/                                                                                                                                                                                                                                                                                                                                                                                                           |

|                              |                                                                                                                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | ios harness was touched, no gui_record* / record_tui_views.sh invoked                                                                                                                                                     |
| <b>Fresh-corpus rotation</b> | §11.4.154 — no prior helixcode-web-* artefacts existed; nothing to rotate                                                                                                                                                 |
| <b>Validator</b>             | DOM-content assertions inside the chromedp harness (per-step real rendered-content checks) + tesseract OCR of the final response frame, with a golden-good/golden-bad analyzer self-validation (§11.4.107(10)/ §11.4.160) |

## Artifacts

| Artifact                        | Path                                                                                       | md5                  |
|---------------------------------|--------------------------------------------------------------------------------------------|----------------------|
| Journey MP4                     | /Volumes/T7/Downloads/Recordings/helixcode-web-llmchat-20260623T131944Z.mp4                | ddf3eaf6b157fa92b3bd |
| Evidence frame (final response) | /Volumes/T7/Downloads/Recordings/helixcode-web-llmchat-20260623T131944Z.evidence_frame.png | 79e5f48385101c8413d3 |

MP4: H.264 / yuv420p / 1280×1256 / 419 frames / ~13.97 s / +faststart.

## Harness (TCC-free, headless Chrome)

helix\_code/tmp/web\_record/main.go (gitignored tmp/; go vet ./tmp/web\_record/ → clean). It: 1. launches headless Chrome (ExecPath = the host Google Chrome, 1280×1400 viewport), 2. navigates to http://localhost:18080/, 3. drives the REAL user journey through the page’s own UI — fill the real login form → click **Sign in** (mints a real server-side JWT) → fill provider/model/prompt → click **Generate**, 4. captures one PNG frame per step from the browser viewport, 5. reads the live DOM at each step and **asserts real rendered content** (a 401 / error page / empty output is a FAIL, never a faked PASS — §11.4.2/ §11.4.107).

Credentials are runtime-generated by the caller and passed via env (WEB\_USER/WEB\_PASS) — **never hardcoded, never logged into evidence** (§11.4.10).

## Server boot (reused HXC-108 approach, §11.4.76 on-demand-infra)

bin/helixcode launched from helix\_code/ (so ./web/frontend resolves) with HELIX\_CONFIG pointing at a **temp config copy** (port: 18080, no tracked-config edit; port 8080 held by an unrelated foreign htCore process, left untouched). Auto-boot detected the already-healthy podman PostgreSQL (: 55432) + Redis (: 56379) and reointed cfg to them;  Database connection established +  Database schema already exists. /health → HTTP 200 in ~2 s. Web UI GET / → real <title>HelixCode – LLM Console</title>; /static/app.js|app.css|logo.png → all 200; GET /api/v1/llm/providers → 8 real providers.

# Durable evidence (committed) — rotation-proof anchors (§11.4.83, HXC-108 audit F2 fix)

The raw corpus (/Volumes/T7/Downloads/Recordings/, §11.4.128/.154-rotatable) is the secondary location. The load-bearing journey MP4 + final-response evidence frame are **copied into the committed tree** so a rotation cannot dangle these citations:

| Committed artifact                                                                   | sha256                        |
|--------------------------------------------------------------------------------------|-------------------------------|
| docs/qa/HXC-108_web/helixcode-web-llmchat-20260623T131944Z.mp4 (h264 1280×1256 419f) | 3d827d223a4611e8faca445207a20 |
| docs/qa/HXC-108_web/helixcode-web-llmchat-20260623T131944Z.evidence_frame.png        | 28e06e45e53b35e9900efcc28d0b0 |

The MP4 ffprobe-verified valid H.264 after copy.

## PER-FEATURE RESULT

### WEB-1. browser LLM Console — load → real login (real JWT) → real authenticated generate (BLUFF-001) — PASS

Real user registered through the live API before driving the UI: POST /api/v1/auth/register → 201 (uuid 322cf640-9bfc-4d55-8053-dcfce93ca6ad, persisted to PostgreSQL); the web UI then logged that user in through its own form.

Harness journey output (DOM read-backs, verbatim):

```
DOM title="HelixCode — LLM Console"  h1="HelixCode LLM Console"
DOM auth-status="authenticated (token stored)"                ← real JWT minted
DOM output="HELIXCODE_WEB_OK4"                                ← real LLM respo
DOM meta="provider=DeepSeek model=deepseek-v4-flash tokens=146 finish=stop"
RESULT: PASS
```

Frames (browser viewport): 01\_console\_loaded, 02\_login\_filled, 03\_authenticated, 04\_generate\_filled, 05\_generate\_requesting, 06\_generate\_response.

**OCR-validated rendered content** (read back from the final frame by tesseract):

```
HelixCode LLM Console
authenticated (token stored)
Provider (optional, e.g.  deepseek
Model (optional)          deepseek-v4-flash
Reply with exactly: HELIXCODE_WEB_OK and then the answer to 2+2
| Response
HELIXCODE_WEB_OK4
```

OCR golden pattern hits: 5/6 (HelixCode, Response, DeepSeek, deepseek-v4-flash, authenticated; tokens sits below the viewport fold in the still frame but is DOM-asserted in the meta line above).

**Analyzer self-validation (§11.4.107(10), the validator provably cannot bluff):** - [golden-good] PASS — the real sentinel HELIXCODE\_WEB\_OK IS present in the rendered frame. - [golden-bad ] PASS — the absent pattern HELIXCODE\_WRONG\_PATTERN\_9999 is correctly reported missing (no hallucinated hit).

**BLUFF-001 cleared:** the model genuinely answered HELIXCODE\_WEB\_OK + 4 via a real DeepSeek call with genuine token accounting (tokens=146, finish=stop) — impossible for a simulated response. Every byte in the #output/#meta regions came from the server's real provider-backed response; the only "no simulation" string OCR sees is the page's own static anti-bluff disclaimer copy (UI chrome describing that output is real), not the LLM reply.

## Real upstream finding (honest, §11.4.6 — not a bluff, surfaced for the conductor)

The web Generate form with an EMPTY model returned **HTTP 502**:

```
generation failed: DeepSeek request failed: DeepSeek API returned status 400
{"error":{"message":"The supported API model names are deepseek-v4-pro or deepseek-v4-flash, but you passed ."}}

```

This is a **real upstream API change**, not a transient: DeepSeek deprecated deepseek-chat (which /api/v1/llm/providers still advertises for the deepseek provider) in favour of deepseek-v4-pro / deepseek-v4-flash. With an explicit valid model (deepseek-v4-flash or deepseek-v4-pro) the same endpoint returns a real content: "4" (verified both via curl AND via the web UI above). The recording therefore drives the journey with an explicit valid model. **Suggested follow-up (conductor):** the server's default-model resolution for DeepSeek (empty model → request body model="") needs to map to a currently-supported model, and the /llm/providers catalog for deepseek should reflect deepseek-v4-\* rather than the deprecated deepseek-chat.

## Teardown (§11.4.14)

SIGINT → server logged 🛑 Shutting down server... /✅ Server exited properly /✅ Database connection pool closed; :18080 freed; foreign htCore on :8080 untouched; podman infra (already healthy before launch, orchestrator-managed) left running.

## Sources verified

- helix\_code/web/frontend/index.html + static/app.js (real login → JWT → authenticated /api/v1/llm/generate, no client-side simulation) — 2026-06-23.
- helix\_code/internal/server/server.go:369-370 (web UI served at / + /static) — 2026-06-23.
- helix\_code/cmd/server/main.go + internal/config/config.go (HELIX\_CONFIG config-path env, §11.4.76 auto-boot) — 2026-06-23.
- Live runs (register/login/generate via curl AND via the web UI) confirmed genuinely working before + during recording (§11.4.6) — 2026-06-23.
- docs/qa/HXC-108\_tui\_server\_recordings\_evidence.md (sibling TUI+server slice; same server-boot approach + analyzer).